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Geochemical Controls on Organic and Inorganic Scale Deposition: Insights from CO₂ Injection in Morrow Fluvial Sandstone

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Abstract

This study investigates the geochemical processes influencing the deposition of organic and inorganic scales during Carbon dioxide (CO₂) injection in hydrocarbon reservoirs, with a focus on Morrow Fluvial Sandstone, Stewart Field, Kansas. CO₂ injection, a widely used enhanced oil recovery (EOR) technique, can lead to paraffin wax and calcium sulfate (CaSO₄) scale deposition, presenting complex flow assurance challenges. Understanding the mechanisms driving these processes is critical for optimizing production strategies and maintaining reservoir integrity. A combination of field data analysis and geochemical modeling was employed to evaluate conditions under which wax and CaSO₄ scales form. Wax appearance temperature (WAT) analysis was used to assess changes in paraffin solubility across temperature ranges, while the effect of CO₂ on light hydrocarbon extraction and WAT was also evaluated. Inorganic scale formation was modeled through geochemical simulations that considered temperature, pressure, and ion concentration variations post-CO₂ injection. Results showed that CO₂ initially lowers WAT and enhances paraffin solubility at low temperatures (e.g., 80°F), but over time, extraction of lighter hydrocarbons raises WAT and promotes solid-phase wax formation, reaching up to 5% near injection wells. In contrast, CaSO₄ scaling occurs from the mixing of incompatible formation waters, leading to concentrations as high as 0.15 mol/L and the precipitation of anhydrite and gypsum near production wells, which significantly reduce permeability. These findings reveal that CO₂-induced geochemical changes exacerbate both organic and inorganic deposition, underscoring the need for integrated management strategies. Unlike previous studies focusing on either wax or scale, this work highlights their interdependent behavior, and the dual challenge

posed during CO₂-EOR. The study contributes to the development of comprehensive flow assurance solutions and more robust production strategies in sandstone reservoirs.

Introduction

Carbon dioxide (CO₂) injection is a widely used enhanced oil recovery (EOR) technique. It can significantly increase hydrocarbon production by several mechanisms, including oil swelling, reducing oil viscosity, and enhancing miscibility between CO₂ and reservoir fluids (Kalyanaraman et al. 2015; Ahmadi et al. 2018). CO₂-EOR has been successfully applied in various sandstone and carbonate reservoirs worldwide, particularly in mature fields where waterflooding has become less effective (Nazari et al. 2017; Longe et al. 2024a). The choice of reservoir for CO₂ injections depends on factors such as depth, temperature, mineralogy, fluid composition, and whether the field has infrastructure that can be adapted for CO₂ handling and injection. The Morrow Fluvial Sandstone in Stewart Field, Kansas, has emerged as a promising candidate for CO₂-EOR due to its favorable geological properties, including an average porosity and permeability. Despite these attractive reservoir characteristics, CO₂ injection operations present complex geochemical challenges that threaten flow assurance, operational efficiency, and long-term reservoir performance (Longe et al. 2024b). Among these challenges are changes in mineral stability, fluid composition, and phase behavior that can disrupt production operations (Alameen et al. 2023). A clear understanding of these mechanisms is necessary for designing effective EOR projects that maintain reservoir productivity while minimizing operational risks and costs, especially in mature fields where economic margins are narrow and operational downtime is costly (Barati Ghahfarokhi and Alhubail 2021; Alameen et al. 2024).

One of the predominant operational challenges encountered during CO₂ injection is the deposition of organic scales, particularly paraffin wax, which can severely restrict flow paths within the reservoir, wellbore, and surface facilities. Wax deposition occurs due to changes in temperature and pressure that alter the solubility behavior of n-alkanes present in crude oil. The wax appearance temperature (WAT) defines the threshold below which paraffin molecules begin to crystallize out of solution, forming solid-phase deposits that reduce effective permeability and obstruct production pathways (Sutton and Roberts 1974). During initial CO₂ injection, oil swelling and extraction of light hydrocarbons lower WAT, thereby enhancing paraffin solubility. However, with prolonged CO₂ exposure, depletion of lighter hydrocarbon components raises WAT, leading to wax crystallization within reservoir pores, tubing, and near-wellbore regions, as observed in field operations and experimental studies (Pedersen et al. 1985; Carnahan 1989). Additionally, CO₂ injection at cooler temperatures compared to the reservoir (e.g., injected at ~80°F vs. reservoir ~125°F) enhances wax precipitation due to localized cooling effects, a critical operational factor influencing paraffin deposition during EOR processes.

In addition to organic deposition, inorganic scale formation, particularly calcium sulfate (CaSO₄) precipitation in the form of gypsum or anhydrite, poses significant risks to reservoir integrity, injectivity, and production sustainability. Geochemical reactions triggered by CO₂ injection result in brine acidification due to carbonic acid formation, altering mineral dissolution and precipitation equilibria (Lagneau et al. 2005). In sandstone reservoirs, CO₂-brine-rock interactions lead to the dissolution of reactive minerals such as feldspars and chlorite, while simultaneously promoting precipitation of minerals like anhydrite, gypsum, and various carbonates (Moore et al. 2005; Al-Khdheawi et al. 2022). Near injection wells, the mixing of incompatible formation waters with injected CO₂ and brine containing high sulfate and calcium concentrations facilitates CaSO₄ scale formation, as temperature and pressure variations reduce its solubility (Hagar et al. 2022). Such precipitation can clog pore throats, reduce injectivity, and obstruct flow paths, ultimately compromising reservoir management strategies (You et al. 2018). This process is further exacerbated by pH reduction and changes in ionic strength due to CO₂ dissolution, which influence mineral

equilibrium constants and reaction kinetics (Hagar and Foroozesh 2021), posing significant flow assurance challenges.

Geochemical rock-brine interactions in sandstone reservoirs are complex due to the heterogeneous mineralogical composition that governs reactivity and dissolution behavior. While quartz, the dominant mineral in sandstone, is relatively inert, associated minerals such as feldspars, chlorite, and carbonates contribute to significant reactivity during CO₂ injection. These interactions influence porosity, permeability, wettability, and interfacial tension, which are critical parameters controlling fluid flow and oil recovery efficiency (Isah et al. 2022). Moreover, injection of CO₂ into sandstone formations decreases brine pH, promoting dissolution of silicate and carbonate minerals and releasing divalent cations such as Ca²⁺ and Mg²⁺. These ions react with carbonate or sulfate species to form insoluble precipitates, such as calcite, magnesite, gypsum, or anhydrite, that block flow channels and reduce reservoir performance (Zhang et al. 2019). The balance between mineral dissolution and precipitation reactions determines net changes in reservoir porosity, permeability, and mechanical integrity. Such reactions are reported by Dontoh et al. to reduce the Young's modulus of rocks including sandstones (Dontoh et al. 2024).

The dual challenges of organic wax deposition and inorganic scaling during CO₂-EOR operations have been investigated extensively in previous studies; however, these have predominantly addressed each phenomenon in isolation. Wax deposition is driven by thermodynamic instability when crude oil cools below its WAT, leading to solid-phase paraffin precipitation that obstructs pore spaces and flow lines. In contrast, scaling is linked to supersaturation of incompatible ions, particularly under temperature, pressure, and compositional changes induced by CO₂ injection, resulting in mineral precipitation such as calcium sulfate (gypsum and anhydrite). While these individual mechanisms are well-documented, the simultaneous occurrence of both deposition processes within the same operational setting remains insufficiently understood. Furthermore, potential synergistic or antagonistic interactions between wax and scale deposition may exacerbate or mitigate flow assurance problems. This knowledge gap limits the ability of operators to design integrated mitigation strategies that effectively manage both challenges in a cost-efficient and operationally feasible manner, especially in fluvial sandstone reservoirs targeted for CO₂-EOR.

Addressing this critical gap requires a holistic investigation that captures how CO₂ injection influences both organic and inorganic deposition mechanisms under reservoir conditions. Such an investigation must integrate fluid thermodynamics, mineral reaction kinetics, temperature and pressure variations, and multiphase flow dynamics to elucidate mechanistic insights into these coupled processes. This study specifically examines the geochemical controls on paraffin wax and calcium sulfate scale deposition within the Morrow Fluvial Sandstone of Stewart Field, Kansas. By integrating field data analysis with advanced compositional and geochemical modeling using CMG STARS and WinProp, the research uncovers how CO₂-induced extraction of light hydrocarbons alters wax precipitation tendencies and how acidified brine chemistry drives mineral scaling. The insights derived from this work aim to inform the development of robust, integrated flow assurance and scale management strategies for CO₂-based EOR projects. Such strategies will enable optimized injection design, targeted chemical treatment, and proactive monitoring to reduce operational risks, improve production efficiency, and enhance the economic viability of CO₂-EOR in similar fluvial sandstone reservoirs.

Methodology

Field Description and Reservoir Characterization

Stewart Field is located in Finney County, Kansas, approximately 12 miles southwest of Garden City, covering an area of ~2400 acres with dimensions of 0.25–0.5 miles wide and 4.5 miles long (Figure 1). Discovered in 1967, production is from the Pennsylvanian Morrow Sandstone Formation at depths of 4755–4766 ft, with most wells completed using 4.5–5.5-inch casing, perforated and stimulated by acid

or diesel treatments. Hydraulic fracturing was introduced in 1990–1991 using gel with sand to enhance productivity (Longe et al. 2024a). The Morrow Formation is divided into upper and lower units, separated by a thin limestone layer, with oil production mainly from the Upper Morrow sandstone. This fluvial channel sandstone was deposited within incised valleys formed by sea-level drops during the Late Mississippian period, resulting in stacked sandstone intervals interbedded with shales. Lithologically, the reservoir consists predominantly of subarkosic quartz-rich sandstone, medium to poorly sorted, with calcite and dolomite cementation and minor clays such as chlorite, illite, and smectite (0–6%). SEM analysis confirms quartz overgrowth, carbonate cementation, and pore throat obstruction by kaolinite and chlorite clay, impacting effective permeability. Average porosity ranges from 13.5% to 16.5%, with permeability between 51 and 130 mD, while the average reservoir temperature is $\sim 125^{\circ}\text{F}$, conditions favorable for CO_2 -EOR applications.

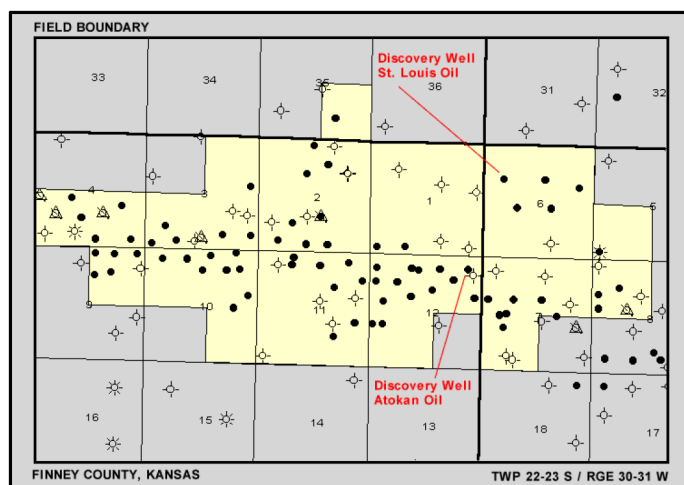


Figure 1—Stewart field, Finney County, Kansas, field boundary map. Located in Atokan Oil and Gas play in the Anadarko basin. Map courtesy of Kansas Geological Survey, (Kansas Geological Survey, 2001).

The Stewart Field Unit (SFU) has been producing since 1967 under an initial depletion drive supported by aquifer influx from the underlying St. Louis and St. Genevieve formations. Waterflooding commenced in 1996 to improve sweep efficiency, and an enhanced oil recovery program using CO_2 injection was initiated in 2012 to further increase recovery. The reservoir has an estimated original oil in place of ~ 33 MMSTB, with net pay thickness ranging from an average of 26 ft to highs of 40 ft. The crude oil has an API gravity of $28\text{--}31.5^{\circ}$ and viscosity of ~ 12 cP, while produced water exhibits high total dissolved solids ($\sim 91,300$ mg/L) and chlorides ($\sim 55,500$ mg/L). Operational challenges include sweep efficiency limitations due to formation heterogeneity and permeability contrasts, as well as scaling and injectivity issues arising from mixing incompatible brines. However, the combination of moderate to high porosity and permeability, quartz-dominated stable mineralogy with carbonate cementation, established infrastructure, and prior CO_2 injection performance makes Stewart Field an excellent candidate for continued CO_2 -based enhanced oil recovery.

Table 1—Stewart Field Unit Data Summary (A. Walton et al. 1997)

Geographic Location	State	Kansas
	County	Finney
General	Number of Wells	43 Producers – 14 dry holes
	Operators	3
Reservoir Characteristics	Formation	Morrow Sand
	Depth to Top	4,764 ft
	Temperature	125 °F
	Original Pressure	1,102 psig
	Average Initial Water Saturation S_{wi}	32.2%
	Area	1,356 Acres
	Original Oil in Place	33 MMSTB
Rock Properties	Lithology	Sandstone (siliciclastic)
	Average Thickness	26 ft
	Average Porosity	16.5%
	Average Permeability	130 md
	Rock Compressibility	$1 \times 10^{-6} \text{ psi}^{-1}$
Fluid Properties	Oil API Gravity	28
	Oil Viscosity (μ) at P_i and T_{res}	12.1 cP
	Initial Solution Gas – Oil Ratio (GOR)	37 SCF/STB
	Gas Specific Gravity (SG)	1.234
	Formation Volume Factor (FVF) at P_{BP}	1.038 RB/STB
	Bubble Point Pressure (P_{BP})	180 psig
	Compressibility at P_i	$5.83 \times 10^{-6} \text{ psi}^{-1}$
	Average Compressibility Above P_{BP}	$7.88 \times 10^{-6} \text{ psi}^{-1}$
Produced Water	Resistivity at 125 F	0.04 Ω -m
	Chlorides	55,500 mg/L
	Total Dissolved Solids (TDS)	91,300 mg/L
	Compressibility at P_i	$3.07 \times 10^{-6} \text{ psi}^{-1}$

Modeling Approach

A static geological model of the Stewart Field Unit (SFU) was constructed to evaluate enhanced oil recovery (EOR) strategies, integrating seismic data, well logs, core analyses, and water chemistry reports. Seismic interpretation defined fault lines and structural horizons, forming the basis of the three-dimensional structural framework, while well log calibration identified key lithological units, depositional trends, and reservoir layering. Core data provided detailed porosity, permeability, and facies distributions, with the final model representing two primary layers: the Morrow Sandstone and the St. Genevieve Formation, between which no significant lateral continuity was observed. As a fluvial reservoir, SFU exhibits substantial heterogeneity manifested as high-porosity and high-permeability channel sands that pinch out laterally into lower-quality facies, impacting sweep efficiency and fluid flow pathways during EOR operations. Figures 2 and 3 illustrate porosity and permeability spatial distributions derived from petrophysical interpretation. The static model also incorporated net-to-gross ratios and lithofacies mapping to enhance rock typing accuracy and flow unit characterization. These details were critical in ensuring realistic reservoir representation before dynamic simulation and evaluating CO₂ injection performance within geological heterogeneity constraints.

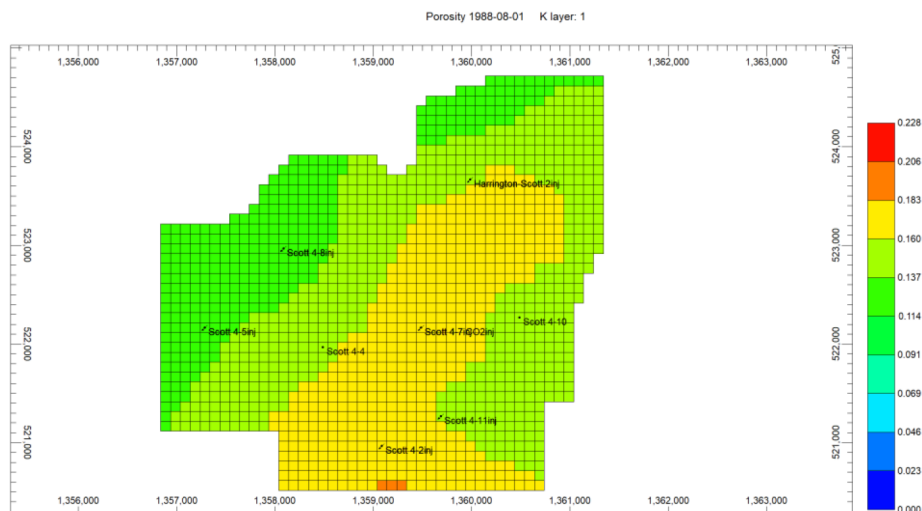


Figure 2—Porosity variation and spatial distribution in SFU

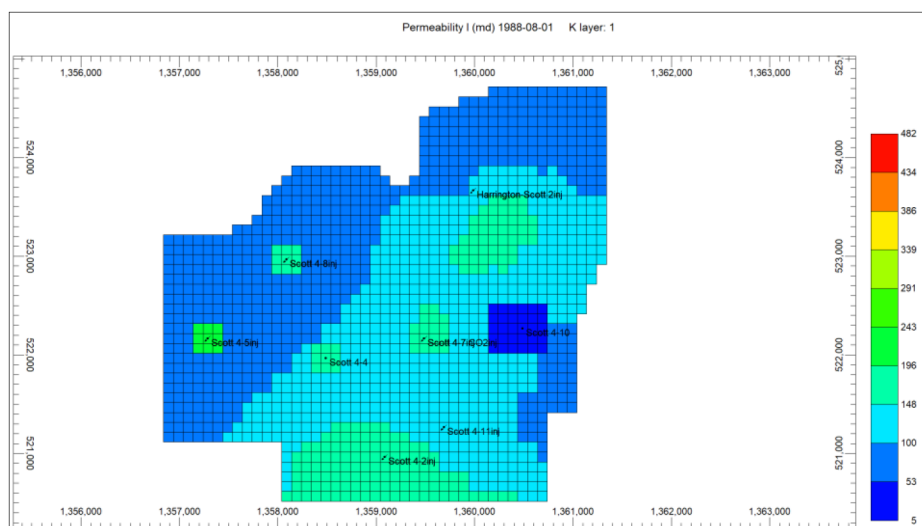


Figure 3—Permeability variation and spatial distribution in SFU

A three-dimensional dynamic sector model was developed using CMG STARS, focusing on the Scott Unit Area (SUA), which includes eight wells, seven producers and one injector. Although the initial full-field model contained approximately 250,000 grid blocks with dimensions of $26 \times 15 \times 3$ ft, it was scaled down to build a sector model that maintained computational efficiency while preserving geological and flow heterogeneity detail. This dynamic model captured historical production phases, including primary depletion (1967–1996), secondary waterflooding (from 1996), and tertiary CO₂ injection initiated in 2017 with conversion of one water injector into a CO₂ injector. Relative permeability curves derived from core test data were implemented (Figure 4), and STARS' advanced thermal and compositional modules allowed pressure and temperature monitoring to evaluate impacts on geochemical processes such as wax precipitation and scale formation. For fluid characterization, CMG WinProp was used to build an equation of state (EOS) model based on laboratory PVT data, lumping the original 36-component system into seven pseudo-components plus CO₂ for computational efficiency while preserving phase behavior (Table A1 in appendix). The Peng-Robinson EOS was tuned to match constant composition expansion (CCE), CO₂ swelling, viscosity, and saturation pressure data, yielding a minimum miscibility pressure (MMP) of 1915 psig, closely matching experimental results (1894 psig). The heavy C₂₀₊ fraction was split into C_{20+A} (liquid)

and C_{20+B} (solid) pseudo-components for wax modeling, with solid mole fraction calculations based on pour point (20°F) and API gravity (31.5°), resulting in a split of 0.2221 mole fraction liquid and 0.0291 mole fraction solid (Table A2 in appendix), ensuring accurate prediction of wax deposition under CO₂ injection conditions.

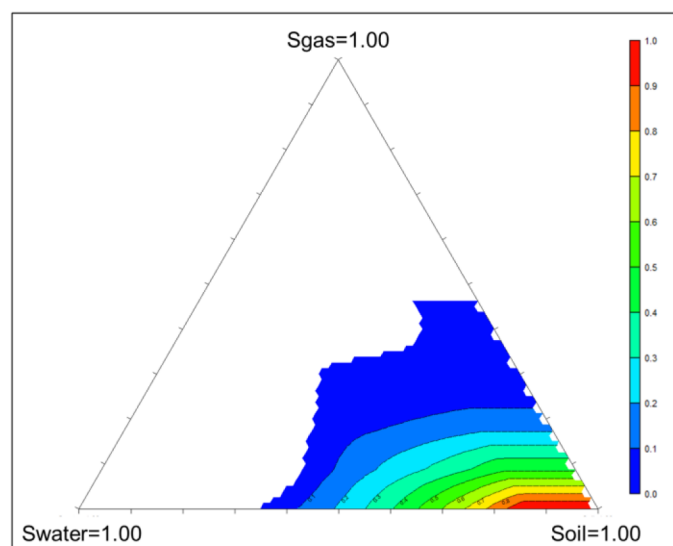


Figure 4—The three phase relative permeability of SFU

Geochemical Modeling

Geochemical modeling was conducted using CMG STARS Thermal & Advanced Processes Simulator to evaluate mineral dissolution and precipitation reactions under CO₂ injection conditions in Stewart Field. The simulator solves transport equations and thermodynamic equilibrium between solid, liquid, and vapor phases, incorporating geochemical reaction formulas for aqueous components (Hagar et al. 2020). Reaction species were categorized as geochemical elements (fundamental components such as Ca²⁺ or Fe²⁺), independent species (species that cannot dissociate further, e.g., Ca²⁺, Mg²⁺), and dependent species (formed through dissociation, complexation, or precipitation reactions). Conservation of mass was ensured through the abundance matrix (A), while chemical equilibria were solved using stoichiometry matrices (S) and kinetic rate equations. Solid mineral reactions were modeled using Arrhenius-based kinetic reaction rates with parameters sourced from the MINTEQ 4 database, including reactive surface area, activation energy, and intrinsic reaction rates. Table 2 summarizes the geochemical components included in the simulation, while Table 3 presents key mineral properties and reaction parameters. Mineral phases modeled included calcite, quartz, and siderite; siderite was used as a proxy for ankerite due to similar thermochemical behaviors despite differences in calcium content. Quartz was assumed chemically inert under simulated conditions, while calcite dissolution and precipitation reactions were critical for evaluating potential calcium carbonate scaling during CO₂ injection. Siderite dissolution reactions represented iron carbonate geochemistry to assess precipitation risks, potential porosity reduction, and permeability impairment near injectors and producers.

Table 2—Geochemical components in STARS.

Variable	Value	Description
ncomp	21	Total number of components in the simulation: Water, H, CO ₂ , SO ₄ , Fe, H ₄ SiO ₄ , Ca, Mg, Na, Cl, CO ₂ , N ₂ -C ₄ , C ₅ -C ₁₈ , C ₉ -C ₁₂ , C ₁₃ -C ₁₆ , C ₁₇ -C ₁₉ , C ₂₀ +A, C ₂₀ +B, Calcite, Quartz, Siderite
n-elem	9	Total number of aqueous elements: Ca ²⁺ , Mg ²⁺ , Na ⁺ , H ₂ O, OH ⁻ , CO ₃ ²⁻ , SO ₄ ²⁻ , H ⁺ , Fe ²⁺
n-aqueous	24	Total number of aqueous species: H ⁺ , CO ₃ ²⁻ , SO ₄ ²⁻ , Fe ²⁺ , H ₄ SiO ₄ , Ca ²⁺ , Mg ²⁺ , Na ⁺ , H ₂ O, CaCO ₃ , OH ⁻ , HCO ₃ ⁻ , H ₂ CO ₃ , MgOH ⁺ , CaOH ⁺ , CaSO ₄ , CaHCO ₃ ⁺ , MgSO ₄ , MgCO ₃ , MgHCO ₃ ⁺ , HSO ₄ ⁻ , NaSO ₄ ⁻
n-solids	3	Total number of solid species. Calcite, Quartz, Siderite
n-sorbs	4	Total number of sorbed species Ca ²⁺ , Mg ²⁺ , Na ⁺ , H ⁺

Table 3—Solid Mineral Reactions in the Reservoir from MINTEQ_V4 Database

Mineral \ Property	Reactive Surface Area m ² /m ³	Activation Energy J/mol	Log 10 of Reaction Rate 1/sec
Quartz	2,650	90,900.1	-13.4
Calcite	2,709.95	23,500	-5.81
Siderite	4,046.67	52,200	-7.53
Chlorite	-	-	-

The aqueous geochemical reactions modeled included water dissociation, carbonate equilibria involving CO₂ hydration to carbonic acid and interconversion between carbonate and bicarbonate ions, as well as calcium carbonate and sulfate precipitation/dissolution equilibria. Magnesium carbonate and sulfate equilibria, and sulfate–bisulfate equilibrium were also included to ensure accurate ion balance and scaling predictions. These reactions were implemented using the MINTEQ 4 chemical database, enabling calculation of species concentrations at reservoir temperature and pressure. Reaction rates were calculated based on reactive surface area, activation energy, and saturation index (Q/K_{eq}), indicating dissolution when <1 and precipitation when >1. Brine composition input was based on Sherman #5 water analysis (Table 4), which revealed high chloride (23,000 ppm), sodium (12,362 ppm), calcium (1,495 ppm), and magnesium (965 ppm) concentrations with a pH of 6.4, conditions conducive to both calcite and calcium sulfate scaling under CO₂ acidification. The final geochemical model quantified mineral reaction impacts on porosity, permeability, and fluid chemistry, providing insights into CO₂–formation water interactions necessary for designing effective flow assurance and scaling mitigation strategies in Stewart Field CO₂ EOR operations.

Table 4—Sherman #5 Water and Composition Analysis.

May 2, 1986, Sherman #5	
Component / Concentration	ppm
Chlorides (Cl ⁻)	23,000
Sodium (Na ⁺)	12,362
Calcium (Ca)	1,495
Magnesium (Mg)	965
Sulphates (SO ₄ ²⁻)	2,070
Bicarbonates (HCO ₃ ⁻)	172
Iron (Fe)	4
Sulphate Reducing Bacteria	27,706
Specific Gravity	1.031
pH	6.4

Results and Discussion

Paraffin / Wax Precipitation Model

The wax precipitation behavior of the Sherman #8 crude oil sample was analyzed over a temperature range from 30 °F to 130 °F at atmospheric pressure (14.7 psia). The modeling results demonstrate a clear thermodynamic response of the crude oil system to temperature variation, particularly the formation and dissolution of paraffin wax as a function of temperature. As shown in Figure 5, the solid wax precipitate weight percentage exhibits a strong inverse relationship with temperature. At 30 °F, wax content reaches its maximum, approximately 5.75 wt%. As the temperature increases, the amount of solid wax declines steadily and becomes negligible beyond 130 °F. This confirms that lower temperatures strongly favor wax precipitation due to the reduced solubility of higher-molecular-weight n-alkanes in the oil phase. Such behavior is critical to consider in CO₂ injection operations, where injected fluids—often cooler than reservoir conditions—can significantly lower local temperatures and trigger wax formation.

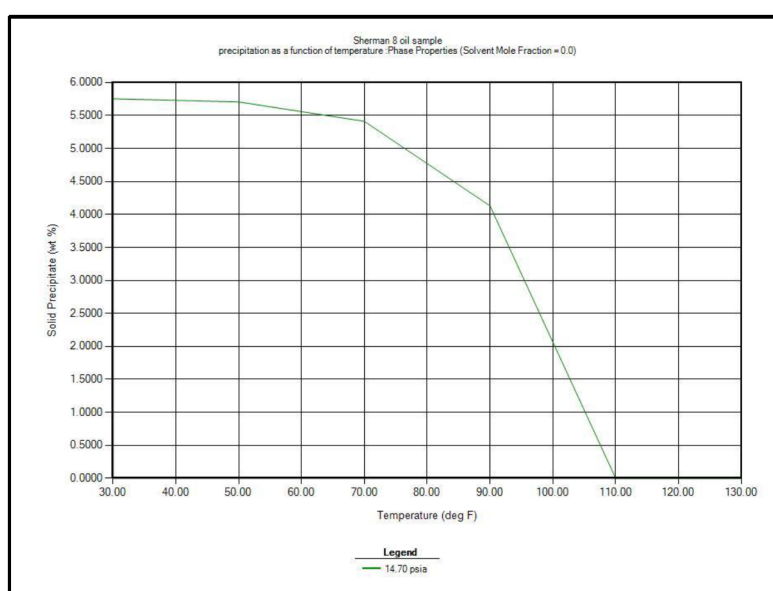


Figure 5—Solid precipitates wt% as a function of temperature.

The impact of CO₂ on wax behavior is further complicated by geochemical interactions. In particular, the acidification of the reservoir through CO₂ dissolution into formation water plays a substantial role in modifying wax precipitation dynamics. CO₂ forms carbonic acid when dissolved in water, reducing the pH of the formation brine. This acidification alters the solubility of n-alkanes and promotes wax crystallization and deposition (Songolzadeh et al. 2021). As illustrated in Figure 6, wax density (specifically the C_{20+B} fraction representing solid paraffinic components) increases as pH decreases. This effect becomes especially pronounced when the pH drops significantly, as CO₂ buffering capacity is overwhelmed during sustained injection. These results indicate that pH-driven geochemical changes are as crucial as thermal effects in understanding wax deposition risks during CO₂-enhanced oil recovery operations.

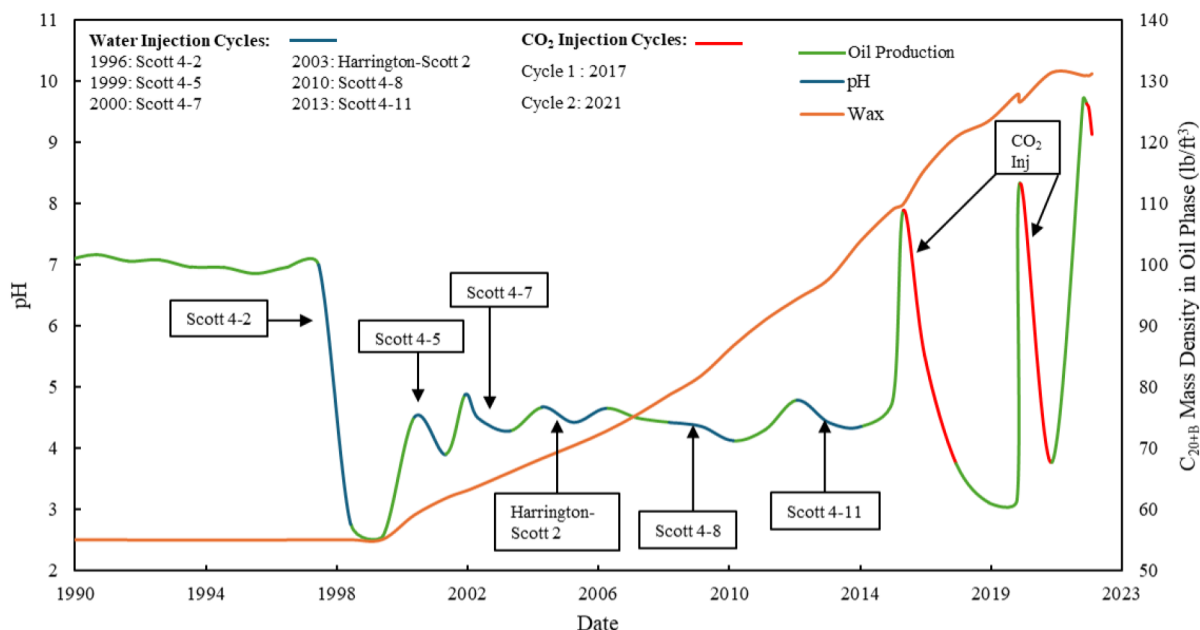


Figure 6—Scott 4-7 pH - wax correlation.

Field-level insights from the Scott 4-7 injection well further support the modeled findings. [Figure 7](#) displays the temperature profile over successive injection cycles. It reveals that wax formation becomes more pronounced over time as reservoir temperature progressively declines due to alternating water and CO₂ injection. Initially, wax formation is minimal; however, with each injection cycle, the cooling effect of the injectants reduces local temperatures, triggering increased wax deposition. This demonstrates how operational practices, especially those involving sub-optimal thermal management, can exacerbate wax-related flow assurance issues. Taken together, the results confirm that temperature decline, acidification from CO₂, and dynamic reservoir conditions all contribute to enhanced paraffin deposition in fluvial sandstone reservoirs.

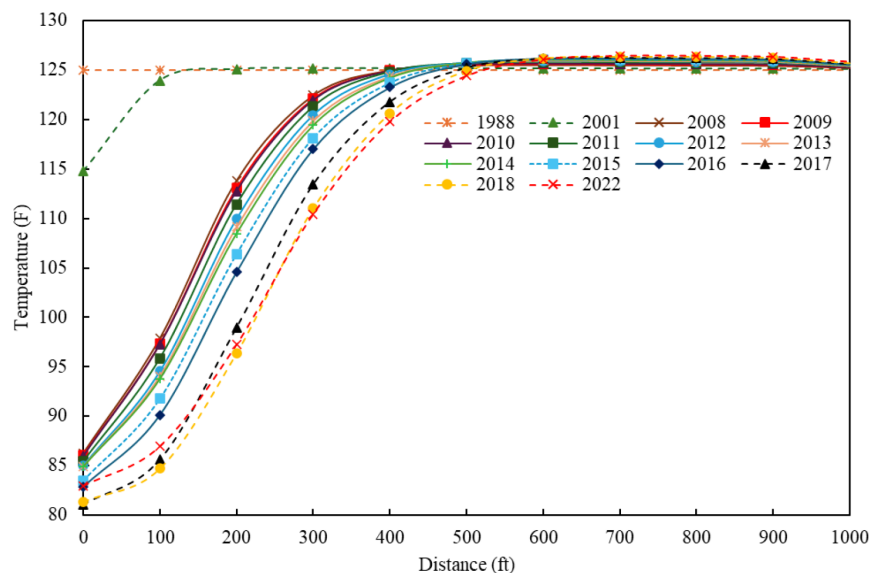


Figure 7—Scott 4-7 injection well temperature profile.

[Figures 8 and 9](#) highlight the influence of thermodynamic conditions on wax (C_{20+B}) density in the Stewart Field Unit. Temperature is shown to exert the most pronounced control on wax phase behavior. As seen in

Figure 8, a progressive decline in reservoir temperature over the years, driven primarily by CO₂ injection, correlates with a steady increase in wax density. The cooling effect of injected CO₂, introduced at ~80 °F compared to the native reservoir temperature of ~125 °F, significantly enhances paraffin precipitation by lowering the wax appearance temperature (WAT). This trend aligns with prior observations that lower temperatures increase the thermodynamic instability of n-alkanes, promoting their crystallization and deposition. The field-level data confirms that wax deposit intensifies during prolonged CO₂ injection cycles, underscoring the operational importance of thermal management in mitigating paraffin-related flow assurance risks.

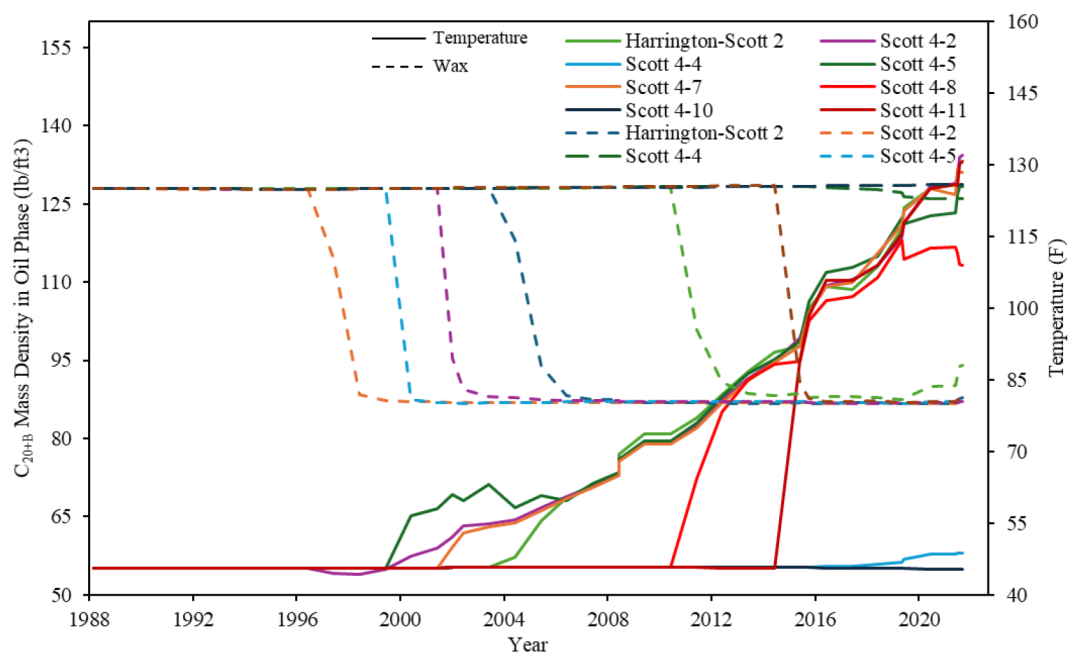


Figure 8—Effect of temperature on wax (C_{20+B}) density

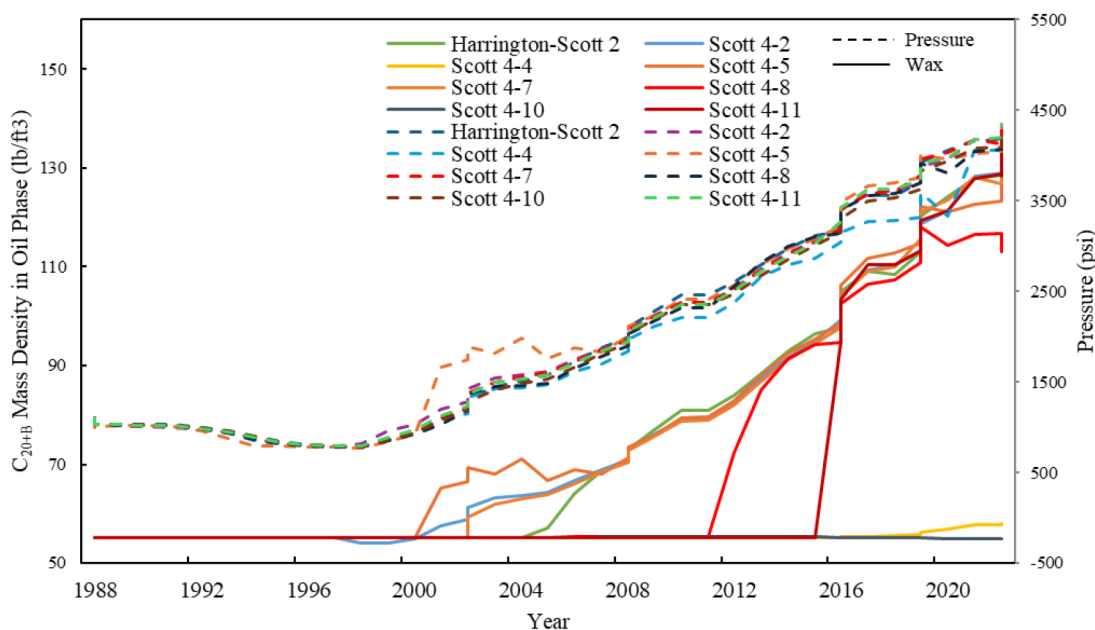


Figure 9—Effect of pressure on wax (C_{20+B}) density

In contrast, Figure 9 illustrates that while pressure effects are less dominant than temperature, they are still non-negligible in influencing wax precipitation. With increasing reservoir pressure, wax density continues to rise gradually, reflecting the reduced solubility of heavier hydrocarbons under higher pressure conditions. This pressure-induced shift in phase behavior lowers the effective WAT and promotes paraffin crystallization within the oil phase. The combined effect of cooling and pressure buildup is particularly evident after 2010, when successive injection cycles caused both lower reservoir temperatures and higher pressures, accelerating wax accumulation across multiple wells. These findings indicate that although temperature is the primary driver, pressure variations under CO₂ injection operations further exacerbate wax deposition, reinforcing the need to incorporate both factors into predictive models and mitigation strategies.

History Matching

The history matching process was essential for calibrating the simulation model to match the observed production and injection data from the Stewart Field Unit (SFU). This step ensures the reliability of the model in capturing historical reservoir behavior and increases confidence in its predictive capability under future operational scenarios (Longe et al. 2025). As illustrated in Figures 10 to 12, history matching was performed for oil production, water production, water injection, CO₂ injection, and CO₂ production across the entire field. The reservoir model focused on the Scott Unit Area (SUA), a smaller sector within SFU, yet several injection wells located on the borders of SUA introduced potential fluid interactions with regions outside the model domain. This cross-boundary fluid migration resulted in slight discrepancies between measured and simulated water production data due to unaccounted water flow from these external wells. To mitigate this, the model leveraged full-field injection allocation data from a prior simulation by Longe et al. (2024), enabling better-informed distribution of water volumes for the history match process.

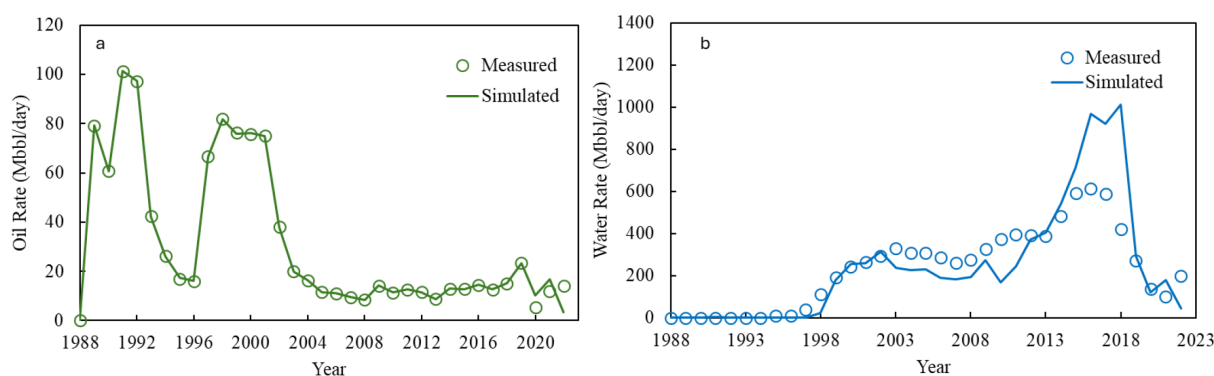


Figure 10—**a-b**—Simulated vs. observed SFU oil (a) and water (b) production rates during history matching.

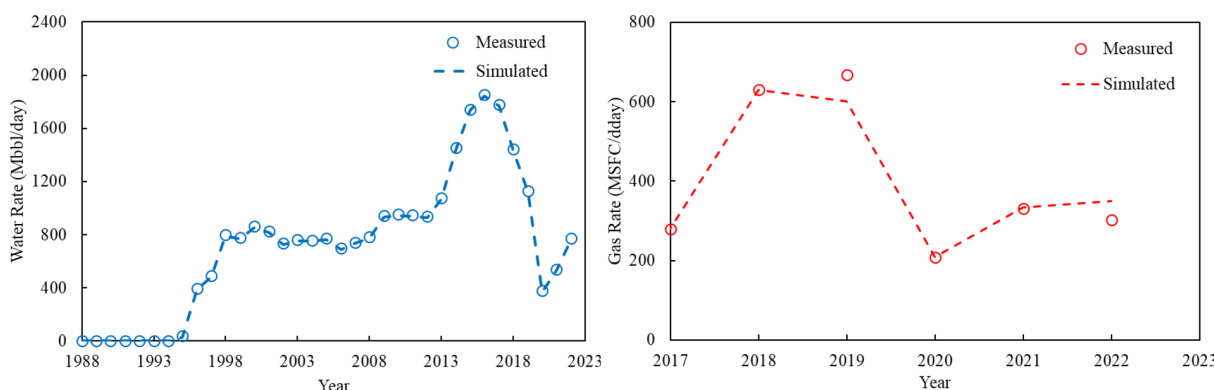


Figure 11—**a-b**— Simulated vs. observed SFU water (a) and CO# (b) injection rates, showing WAG cycle effects.

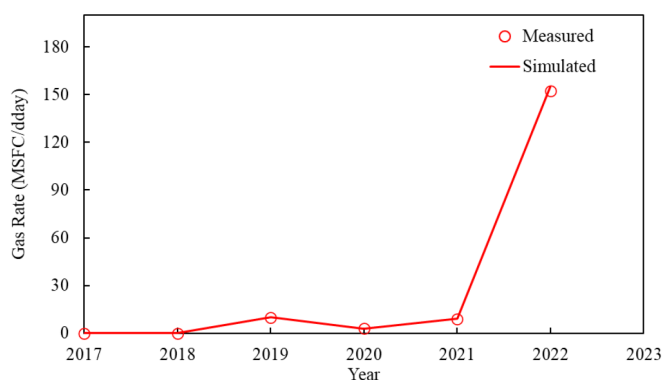


Figure 12—Simulated vs. observed SFU CO₂ production rates, indicating breakthrough and recycling trends.

The simulated CO₂ injection trends shown in Figure 11b highlight two distinct WAG (Water-Alternating-Gas) cycles initiated in 2017 and 2021. The first cycle displayed a steady increase in injection rates, reaching approximately 700 MSCF/day in 2018, followed by a decline during alternating water slugs and a pause in CO₂ injection by 2020. The second WAG cycle commenced in 2021 and demonstrated a more stable injection pattern, plateauing around 400 MSCF/day by 2022. These cycles were modeled accurately, and the resulting simulation trends closely follow field measurements, reflecting effective reservoir pressure maintenance and improved sweep efficiency. This level of agreement between modeled and actual data supports the validity of the WAG strategy as an effective tertiary recovery technique in SFU. Figure 12 further illustrates CO₂ production behavior, with relatively low values between 2017 and 2020 and a sharp increase in 2021, suggesting CO₂ breakthrough and recycling, which matches the observed field trend.

Finally, the overall performance of oil recovery strategies is reflected in Figure 13, where cumulative oil production trends are traced through the primary, secondary, and tertiary recovery phases. Primary production, which began in the 1980s, was followed by a significant increase during the secondary recovery phase—specifically waterflooding—in the late 1990s. Peak oil production reached approximately 1,000 barrels per day by 2005. The introduction of CO₂-EOR in 2017 ushered in the tertiary recovery phase, helping to sustain oil rates despite natural reservoir decline. The gradual upward trend in cumulative production from 2018 onward demonstrates the effectiveness of CO₂ injection in mobilizing residual oil and extending the productive life of the field. The alignment between measured and simulated data across all variables reinforces the robustness of the model calibration and its suitability for subsequent scenario analysis and optimization of EOR strategies.

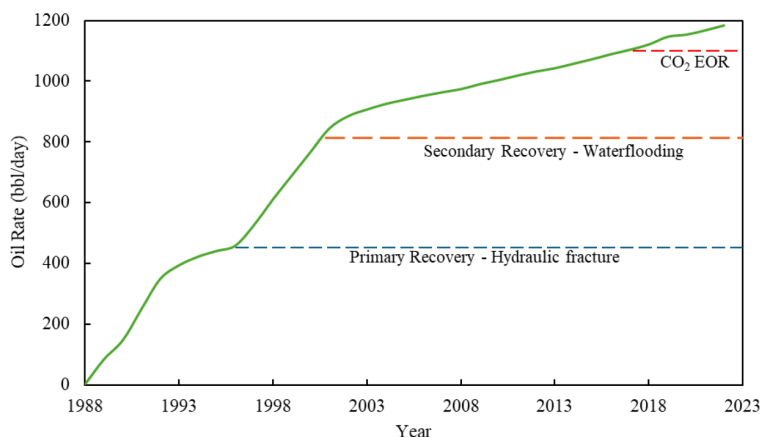


Figure 13—Cumulative SFU oil production across primary, secondary, and tertiary recovery phases.

Geochemical Reactions

The geochemical modeling results reveal the profound impact of CO₂ injection on the aqueous chemistry and mineral stability within the SFU reservoir, particularly in the Scott Unit. As CO₂ dissolves in brine, it forms carbonic acid, lowering the pH and disrupting long-established fluid–rock equilibria. This acidification affects both mineral solubility and ionic speciation, driving a series of interconnected geochemical reactions. Tables 5–7 detail the dominant aqueous reactions, mineral precipitation pathways, and ion exchange processes governing these transformations. The thermodynamic favorability of most aqueous reactions (12 out of 15 have positive log Keq values) suggests strong tendencies toward the generation of bicarbonate, carbonate, and sulfate species under reservoir conditions, all of which can directly or indirectly contribute to scale formation. Meanwhile, the negative Ksp values for minerals such as calcite and siderite indicate low solubility under current conditions, yet localized changes in pH and temperature due to injection cycles can still induce precipitation if supersaturation thresholds are crossed.

Table 5—Aqueous Reaction and Equilibrium Constant (K^{eq}) of geo-elements in SFU

Reaction	Equilibrium Coefficient Equation (species concentrations)	Equilibrium Coefficient Value (Logarithm)
$(CO_3^{2-}) + (Ca^{2+}) \leftrightarrow CaCO_3$	$K_1^{eq} = \frac{[CaCO_3]}{[CO_3^{2-}][Ca^{2+}]}$	3.2
$H_2O \leftrightarrow (OH^-) + (H^+)$	$K_2^{eq} = [H^+][OH^-]$	-13.997
$(H^+) + (CO_3^{2-}) \leftrightarrow (HCO_3^-)$	$K_3^{eq} = \frac{[HCO_3^-]}{[H^+][CO_3^{2-}]}$	10.329
$2(H^+) + (CO_3^{2-}) \leftrightarrow (H_2CO_3)$	$K_4^{eq} = \frac{[H_2CO_3]}{[H^+]^2[CO_3^{2-}]}$	16.681
$(Mg^{2+}) + H_2O \leftrightarrow (MgOH^+) + (H^+)$	$K_5^{eq} = \frac{[MgOH^+][H^+]}{[Mg^{2+}]}$	-11.397
$(Ca^{2+}) + H_2O \leftrightarrow (CaOH^+) + (H^+)$	$K_6^{eq} = \frac{[CaOH^+][H^+]}{[Ca^{2+}]}$	-12.697
$(Ca^{2+}) + (SO_4^{2-}) \leftrightarrow (CaSO_4)$	$K_7^{eq} = \frac{[CaSO_4]}{[Ca^{2+}][SO_4^{2-}]}$	2.36
$(Ca^{2+}) + (H^+) + (CO_3^{2-}) \leftrightarrow (CaHCO_3^+)$	$K_8^{eq} = \frac{[CaHCO_3^+]}{[Ca^{2+}][H^+][CO_3^{2-}]}$	11.599
$(Mg^{2+}) + (SO_4^{2-}) \leftrightarrow (MgSO_4)$	$K_9^{eq} = \frac{[MgSO_4]}{[Mg^{2+}][SO_4^{2-}]}$	2.26
$(Mg^{2+}) + (CO_3^{2-}) \leftrightarrow MgCO_3$	$K_{10}^{eq} = \frac{[MgCO_3]}{[Mg^{2+}][CO_3^{2-}]}$	2.29
$(Mg^{2+}) + (H^+) + (CO_3^{2-}) \leftrightarrow (MgHCO_3^+)$	$K_{11}^{eq} = \frac{[MgHCO_3^+]}{[Mg^{2+}][H^+][CO_3^{2-}]}$	11.339
$(H^+) + (SO_4^{2-}) \leftrightarrow (HSO_4^-)$	$K_{12}^{eq} = \frac{[HSO_4^-]}{[H^+][SO_4^{2-}]}$	1.99
$(Na^+) + (SO_4^{2-}) \leftrightarrow (NaSO_4^-)$	$K_{13}^{eq} = \frac{[NaSO_4^-]}{[Na^+][SO_4^{2-}]}$	429.923
$(Na^+) + (CO_3^{2-}) \leftrightarrow (NaCO_3^-)$	$K_{14}^{eq} = \frac{[NaCO_3^-]}{[Na^+][CO_3^{2-}]}$	1.27
$(Na^+) + (H^+) + (CO_3^{2-}) \leftrightarrow (NaHCO_3)$	$K_{15}^{eq} = \frac{[NaHCO_3]}{[Na^+][H^+][CO_3^{2-}]}$	10.079

Table 6—Precipitation Reactions and Solubility Constraints (K^{sp}) of SFU geo-elements.

Reaction	Solubility Product (species concentration)	Solubility Product Constant (Logarithm)
$(CO_3^{2-}) + (Ca^{2+}) \leftrightarrow \text{Calcite } (CaCO_3)$	$K_1^{sp} = [Ca^{2+}][CO_3^{2-}]$	-8
$(H_4SiO_4) \leftrightarrow \text{Quartz } (SiO_2) + H_2O$	$K_2^{sp} = [H_4SiO_4]$	-4
$(Fe^{2+}) + (CO_3^{2-}) \leftrightarrow \text{Siderite } (FeCO_3)$	$K_4^{sp} = [FeCO_3]$	-10.24

Table 7—Ion-Exchange Reactions and the Sorbed Stoichiometry Coefficients (K^{exm}).

Reaction	Exchange Coefficient Equation (species concentrations)	Exchange Coefficient Value (Logarithm)
$2\overline{Na}^+ + Ca^{2+} \leftrightarrow 2Na^+ + \overline{Ca}^{2+}$	$K_1^{exm} = \frac{[Ca^{2+}][2Na^+]}{[2\overline{Na}^+][Ca^{2+}]}$	-6.43
$\overline{H}^+ + Na^{2+} \leftrightarrow H^+ + \overline{Na}^+$	$K_2^{exm} = \frac{[Na^+][H^+]}{[\overline{H}^+][Na^+]}$	0.899
$2\overline{Mg}^+ + Ca^{2+} \leftrightarrow 2Mg^+ + \overline{Ca}^{2+}$	$K_3^{exm} = \frac{[Ca^{2+}][2Mg^+]}{[2\overline{Mg}^+][Ca^{2+}]}$	0.716

Among the geochemical reactions, $CaSO_4$ (calcium sulfate) emerged as the primary scale-forming compound of concern. In contrast, $CaCO_3$ (calcite) and $MgCO_3$ (magnesium carbonate) were shown to remain undersaturated throughout injection cycles, particularly due to the acidic environment induced by CO_2 -brine interactions. This is visually confirmed in Figures 14 a-c, which show that aqueous concentrations of $CaCO_3$ and $MgCO_3$ remain below their respective solubility limits, while $CaSO_4$ exceeds its threshold significantly, with concentrations reaching up to 0.0086 mol/L against a solubility limit of 0.0049 mol/L. This finding is crucial as it clearly delineates the conditions under which inorganic scale poses a tangible threat to injectivity, wellbore integrity, and near-wellbore permeability. Furthermore, these results align well with observed operational challenges in similar CO_2 -EOR projects, where sulfate scale has been a persistent issue due to water incompatibility and thermal shifts.

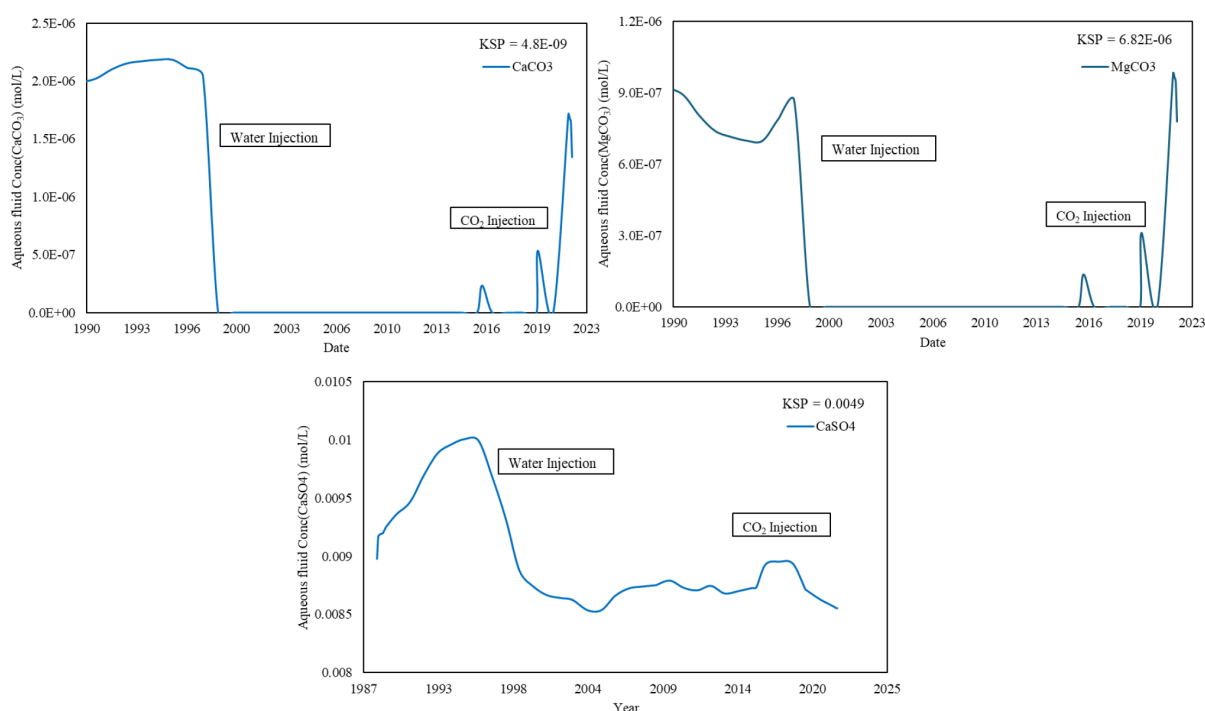


Figure 14—Aqueous concentrations of $CaCO_3$ (a), $MgCO_3$ (b), and $CaSO_4$ (c) in the Scott 4-7 Injection Well during CO_2 -WAG cycles, showing scale formation potential relative to solubility thresholds.

Thermal behavior analysis reveals that temperature fluctuations during water and CO_2 injection cycles further control the precipitation dynamics of $CaSO_4$. Initially, water injection lowers reservoir temperature to $\sim 80^\circ F$, suppressing solubility and favoring precipitation. Subsequent CO_2 injection raises the temperature to $\sim 90^\circ F$ due to exothermic reactions (e.g., carbonic acid and bicarbonate formation shown in reactions 35 and 36), leading to partial redissolution or stabilization of the brine chemistry. This temperature-pH interplay is significant because it confirms that CO_2 injection has a dual chemical-thermal effect on the

reservoir, which must be considered during scale management planning. Figure 15 further reinforces this trend by tracking CaSO_4 concentration over time in the Scott 4-7 well. The pattern shows initial dilution during secondary recovery, followed by renewed concentration growth post- CO_2 injection. The resurgence in sulfate levels post-2017 demonstrates how miscible gas flooding not only mobilizes hydrocarbons but also reshapes chemical gradients within the reservoir. The CO_2 -WAG cycles result in temporary pH drops, enhancing initial solubility, but as CO_2 stabilizes, precipitation dominates due to increased calcium and sulfate availability in the brine.

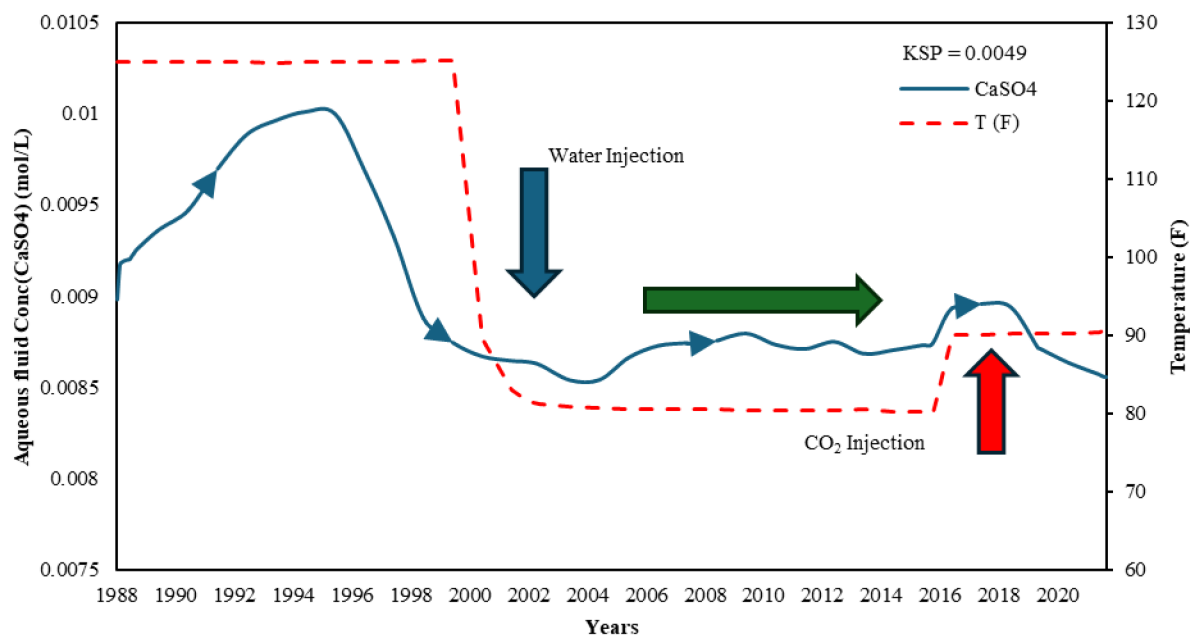
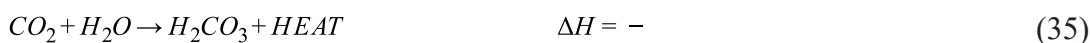


Figure 15—Scott 4-7 Injection well CaSO_4 aqueous fluid concentration.



To deepen the analysis, Figure 16 was used to assess the polymorphic behavior of calcium sulfate. The results show that at varying temperatures within the SFU from 125°F during primary production, to 80°F during waterflooding, and up to 90°F during CO_2 -WAG different hydrate forms of CaSO_4 precipitate. While anhydrite and gypsum dominate under reservoir-relevant conditions, hemihydrate formation is thermodynamically unlikely. This differentiation is vital because each polymorph exhibits distinct solubility, morphology, and adherence properties, directly affecting scale removal strategies and flow assurance interventions. These polymorphic behaviors, combined with dynamic changes in brine composition and thermal shifts from CO_2 injection, create a complex geochemical system. In summary, the geochemical simulation results substantiate that CO_2 injection into the SFU reservoir triggers a temperature-sensitive chemical environment, with calcium sulfate posing the most significant scaling risk. The strong alignment between modeled concentration trends and thermodynamic predictions boosts confidence in the reservoir model's geochemical fidelity. These insights underscore the importance of coupling geochemical diagnostics with thermal and operational modeling to ensure effective scale control. This knowledge provides a foundation for optimizing inhibitor selection, tailoring WAG schedules, and improving injectivity and sustainability in CO_2 -EOR operations in sandstone reservoirs.

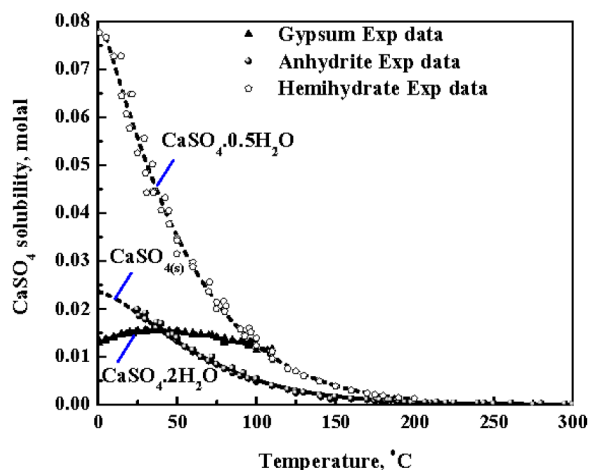


Figure 16—Solubility of CaSO_4 polymorphs (Azimi & Papangelakis, 2008).

Reservoir Characteristics

Incorporating geochemical processes into the reservoir simulation model significantly impacted the average reservoir pressure and revealed the critical role that in-situ chemical reactions play in altering reservoir performance. Over the 34-year simulation window from initial production through various secondary and tertiary recovery stages, the reservoir behavior differed notably between the geochemistry-integrated case and the base model. Initially, before significant precipitation occurred, the average reservoir pressure was around 1,000 psi, providing a key baseline for evaluating later deviations. However, following the onset of paraffin (wax) precipitation, a marked pressure buildup was observed. Particularly, in the geochemical model, bottom hole pressure increased dramatically from 1,600 psi to 4,798 psi after 2006, the period when the waterfront from successive injection cycles reached the production wells. In contrast, the base model showed a smaller pressure increase, reaching only 4,222 psi over the same period. This divergence can be directly linked to the permeability impairment caused by solid wax accumulation in the pore network, which creates flow resistance and necessitates increased pressure to sustain production rates. These changes illustrate how organic deposition can significantly alter flow dynamics, particularly under the cooler reservoir temperatures and acidic pH conditions induced by CO_2 injection cycles.

Figure 17 shows the effect of geochemistry and the difference in bottom hole pressure of a producer and an injector. The influence of inorganic scaling, specifically calcium sulfate precipitation, further exacerbated the pressure increase and highlighted the importance of accounting for multiphase geochemical interactions. As aqueous calcium and sulfate ions exceed their solubility product during post-injection stabilization, the formation of solid CaSO_4 deposits contributes additional obstruction within pore spaces. When both wax and calcium sulfate scale are factored into the simulation, the cumulative impact results in an even higher average reservoir pressure reaching 4,811 psi. This pressure escalation poses operational challenges for maintaining injectivity and controlling production rates, especially under water-alternating-gas (WAG) conditions where frequent thermal and compositional shifts accelerate precipitation kinetics. The findings emphasize the necessity of including geochemical equilibrium reactions in CO_2 -EOR modeling to avoid underestimating pressure buildup and flow impairment risks. Without this integration, traditional reservoir models may misrepresent reservoir deliverability and delay detection of flow assurance issues. Therefore, proactive strategies such as adjusting injection temperature, applying chemical inhibitors, and optimizing WAG timing have become essential for sustaining mobility and enhancing oil recovery. Figure 18 presents a comparative visualization of average reservoir pressure under scenarios with and without geochemical integration, highlighting the stark operational implications of neglecting subsurface chemical behavior.

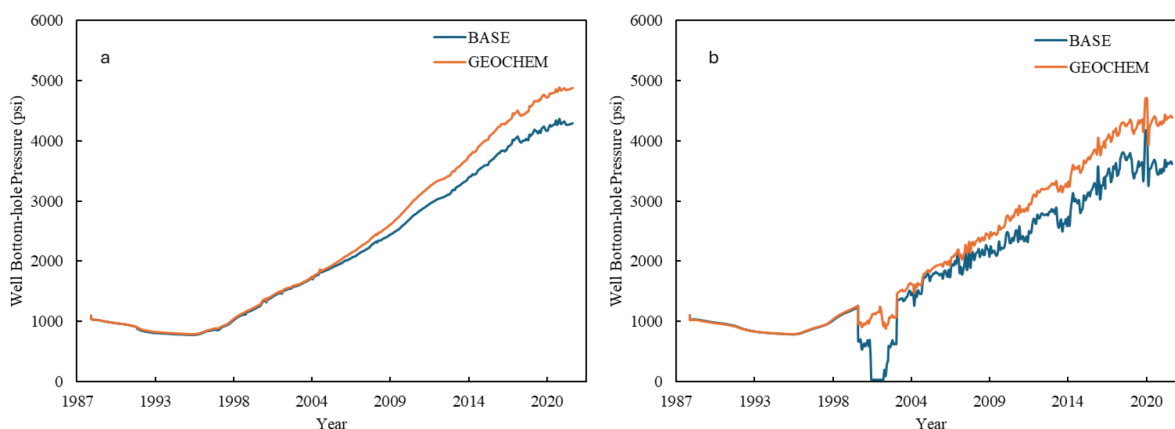


Figure 17—The effect of geochemistry on Scott 4-7 Injection (a) & Scott 4-10 Production Well (b) Bottom Hole Pressure Calculations. BHP increases when geochemistry is included likely due to clogging of pores and buildup of scales caused by precipitation of organic and inorganic solids.

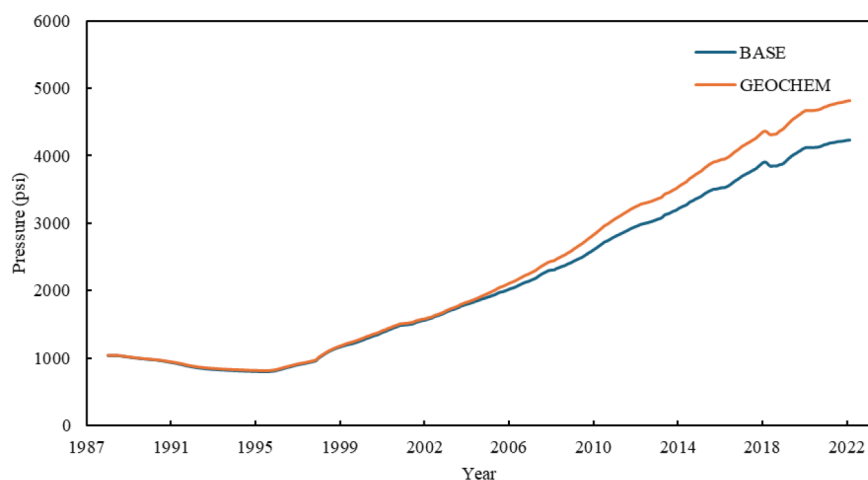


Figure 18—Average reservoir pressure.

Conclusions

This study provides new insights into the coupled geochemical and thermodynamic processes driving paraffin wax and calcium sulfate scale deposition during CO₂-enhanced oil recovery (EOR) in the Stewart Field Unit, Kansas. The main findings are:

- Wax formation was most severe near CO₂ injection wells where temperatures dropped to ~80 °F, resulting in solid-phase wax precipitation up to 5 wt%.
- Initial CO₂ injection lowered the wax appearance temperature (WAT) by enhancing paraffin solubility, but prolonged exposure extracted lighter hydrocarbons, increased WAT, and promoted wax crystallization and deposition.
- Calcium sulfate scaling intensified under CO₂ injection, with concentrations reaching 0.15 mol/L and precipitation of anhydrite and gypsum polymorphs due to brine acidification, temperature variations, and mixing of incompatible waters.
- Geochemical modeling predicted lower porosity and higher reservoir pressure buildup compared to the base case, with greater impairment under high-compaction scenarios due to mineral precipitation, wax plugging, and mechanical pore collapse.
- CO₂ dissolution mobilized cations that precipitated with anions to form solid phases, reducing flow capacity and reservoir performance.

- Coupled modeling of geochemical and thermodynamic processes is critical for accurate forecasting, effective flow assurance management, and improving the long-term economics of CO₂-EOR operations.

References

- A. Walton, Don W. Green, G. Paul Whillite, et al (1997) Improved Oil Recovery in Fluvial Dominated Deltaic Reservoirs of Kansas - Near-Term. <https://doi.org/10.2172/1667>
- Ahmadi MA, Zendehboudi S, James LA (2018) Hybrid connectionist model determines CO₂-oil swelling factor. *Pet Sci* **15**: 591–604. <https://doi.org/10.1007/S12182-018-0230-5>,
- Alameen MB, Elraies KA, Almansour A, et al (2023) Quartz Hydrolysis Analysis of Pure Quartz for Enhanced Oil Production; Influence of Time, pH, and Salinity on Hydrolysis. *Journal of Advanced Research in Fluid Mechanics and Thermal Sciences* **111**: 99–106. <https://doi.org/10.37934/ARFMTS.111.2.99106>
- Alameen MB, Elraies KA, Mohyaldinn M, et al (2024) Reducing the Sand Production from Semi-Consolidated Sandstone Formation by Mitigating the Silica Dissolution Factor During Water Breakthrough. *International Petroleum Technology Conference, IPTC 2024* 12–14. <https://doi.org/10.2523/IPTC-23402-EA>
- Al-Khdeewi EA, Mahdi DS, Hasan MM (2022) Reservoir scale CO₂-water-rock interactions and geochemical evolution of sandstone reservoirs due to CO₂geo-storage process. *AIP Conf Proc* 2443:. <https://doi.org/10.1063/5.0091933/2823524>
- Barati Ghahfarokhi Reza, Alhubail Mmakki (2021) Unconventional hydrocarbon resources : techniques for reservoir engineering analysis. 584
- Carnahan NF (1989) Paraffin deposition in petroleum production. *Journal of Petroleum Technology; (USA)* **41**: 10: <https://doi.org/10.2118/19895-PA>
- Dontoh WH, Mews K, Ghanbarian B, et al (2024) Measuring Variations in Rock Mechanical Properties Due to CO₂ Injection: Application of Nano-Indentation. Society of Petroleum Engineers - *ADIPEC 2024*. <https://doi.org/10.2118/222758-MS>
- Hagar HS, Foroozesh J, Zivar D, et al (2020) Simulation of Hydrogen Sulfide Generation in Oil and Gas Geological Formations. *2020 International Conference on Computational Intelligence, ICCI 2020* 121–125. <https://doi.org/10.1109/ICCI51257.2020.9247695>
- Hagar HS, Foroozesh J (2021) Effect of CO₂ and salinity on microbial hydrogen sulphide generation in hydrocarbon reservoirs. *J Nat Gas Sci Eng* **96**: 104288. <https://doi.org/10.1016/j.jngse.2021.104288>
- Hagar HS, Foroozesh J, Kumar S, et al(2022) Microbial H₂S generation in hydrocarbon reservoirs: Analysis of mechanisms and recent remediation technologies. *J Nat Gas Sci Eng* 104729
- Isah A, Arif M, Hassan A, et al(2022) Fluid–rock interactions and its implications on EOR: Critical analysis, experimental techniques and knowledge gaps. *Energy Reports* **8**: 6355–6395. <https://doi.org/10.1016/J.EGYR.2022.04.071>
- Kalyanaraman N, Arnold C, Gupta A, et al(2015) Stability Improvement of CO₂ Foam for Enhanced Oil Recovery Applications Using Polyelectrolytes and Polyelectrolyte Complex Nanoparticles. Society of Petroleum Engineers - *SPE Asia Pacific Enhanced Oil Recovery Conference, EORC 2015* 878–896. <https://doi.org/10.2118/174650-MS>
- Lagneau V, Pipart A, Catalette H (2005) Reactive Transportmodelling and Long Term Behaviour of CO₂ Sequestration in Saline Aquifers. *Oil & Gas Science and Technology* **60**: 231–247. <https://doi.org/10.2516/OGST:2005014>
- Longe P, Molomjav S, Barati R, et al(2024a) Field-Scale Simulations of Water-Alternating-Gas Injection in Morrowan Fluvial Sandstones of Stewart Field, Kansas, Using Captured CO₂ from an Ethanol Plant. *International Petroleum Technology Conference, IPTC 2024*. <https://doi.org/10.2523/IPTC-24517-MS>
- Longe P, Molomjav S, Tsau JS, et al(2025) Techno-economic evaluation of CO₂-EOR and carbon storage in a shallow incised fluvial reservoir using captured-CO₂ from an ethanol plant. *Geoenery Science and Engineering* **246**: 213559. <https://doi.org/10.1016/J.GEOEN.2024.213559>
- Longe PO, Tsau JS, Musgrove S, et al(2024b) An Overview of Stewart Field Unit Project: A Field Case Study of CO₂ Capture, Utilization, and Storage. Society of Petroleum Engineers - *SPE Energy Transition Symposium, ETS 2024*. <https://doi.org/10.2118/221422-MS>
- Moore J, Adams M, Allis R, et al(2005) Mineralogical and geochemical consequences of the long-term presence of CO₂ in natural reservoirs: An example from the Springerville–St. Johns Field, Arizona, and New Mexico, U.S.A. *Chem Geol* **217**: 365–385. <https://doi.org/10.1016/J.CHEMGEO.2004.12.019>
- Nazari N, Tsau JS, Barati R (2017) CO₂ Foam Stability Improvement Using Polyelectrolyte Complex Nanoparticles Prepared in Produced Water. *Energies* 2017, Vol **10**, Page 516 10:516. <https://doi.org/10.3390/EN10040516>
- Pedersen KS, Thomassen P, Fredenslund A (1985) Thermodynamics of Petroleum Mixtures Containing Heavy Hydrocarbons. 3. Efficient Flash Calculation Procedures Using the SRK Equation of State. *Industrial and Engineering Chemistry Process Design and Development* **24**: 948–954.

- https://doi.org/10.1021/I200031A009/ASSET/I200031A009.FP.PNG_V03
- Songolzadeh R, Shahbazi K, Madani M (2021) Modeling n-alkane solubility in supercritical CO₂ via intelligent methods. *J Pet Explor Prod Technol* **11**: 279–287. <https://doi.org/10.1007/S13202-020-01016-X/TABLES/2>
- Sutton GD, Roberts LD (1974) Paraffin Precipitation During Fracture Stimulation. *Journal of Petroleum Technology* **26**: 997–1004. <https://doi.org/10.2118/4411-PA>
- You L, Wang Z, Kang Y, et al(2018) Experimental investigation of porosity and permeability change caused by salting out in tight sandstone gas reservoirs. *Journal of Natural Gas Geoscience* **3**: 347–352. <https://doi.org/10.1016/J.JNGGS.2018.12.003>
- Zhang L, Wang Y, Miao X, et al(2019) Geochemistry in geologic CO₂ utilization and storage: A brief review. *Advances in Geo-Energy Research* **3**: 304–313. <https://doi.org/10.26804/AGER.2019.03.08>

Appendix A

Table A1—WinProp lumping scheme.

Pseudo-component	Category
CO ₂	Carbon Dioxide injectant
C ₁ - C ₄	Light alkanes (including N ₂)
C ₅ - C ₈	Medium alkanes
C ₉ - C ₁₂	Heavy alkanes
C ₁₃ - C ₁₆	Light aromatics
C ₁₇ - C ₁₉	Medium aromatics
C ₂₀₊	Heavier hydrocarbons
C _{20+A}	Heavier hydrocarbons (liquid)

Table A2—SFU Fluid Compositional Analysis.

Components	Mole fraction	MW (g/mol)	SG	Critical pressure (atm)	Critical temperature (°F)	Acentric factor	T _b (°F)
CO ₂	0.003	44.01	0.818	72.8	304.2	0.225	-109.2
N ₂ -C ₄	0.137	42.51	0.502	42.1	351.8	0.131	3.996
C ₅ -C ₁₈	0.204	93.48	0.712	31.1	533.1	0.304	196.4
C ₉ -C ₁₂	0.194	139.3	0.786	24.3	630.7	0.454	347.9
C ₁₃ -C ₁₆	0.138	195.6	0.829	18.9	706.8	0.618	489.6
C ₁₇ -C ₁₉	0.072	249.8	0.856	15.7	759.9	0.757	594.1
C _{20+A}	0.022	381.3	0.903	10.6	864.4	1.049	820.8
C _{20+B}	0.029	381.3	0.903	10.6	864.4	1.049	820.8